FORM HS-SA · REV. 5 POST-STASIS REFRESHER · SUPPLEMENT A POST AT EVERY PERMIT BOARD & ENTRY HATCH

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Safety & Health

THE AIR CAN KILL YOU.

CONFINED SPACES LOOK HARMLESS. THE DANGER IS THE AIR INSIDE, AND YOU CAN'T SEE IT, SMELL IT OR FEEL IT.

WHAT COUNTS AS A CONFINED SPACE

ALL THREE = CONFINED SPACE

This is **Supplement A (HS-SA): Confined Spaces**. Complete it, and hold the badge endorsement for your role (entrant, attendant or entry supervisor), before you take part in any entry.

1
BIG ENOUGH TO GET INTO

Your whole body can enter and do work.

2
HARD TO GET OUT OF

Limited ways in or out: a hatch, a manway, a ladder, a crawl.

3
NOT BUILT FOR PEOPLE TO STAY IN

Nobody is meant to work there all shift.

It becomes a **permit-required space** if it has, or could have, a dangerous atmosphere, something that could engulf you (sand, grain, liquid, polymer), a shape that could trap or suffocate you, or any other serious hazard. **On this site, treat every confined space as permit-required (site rule)** until the entry supervisor signs a written certificate at the hatch that every hazard has been removed without anyone going in. If anything changes, it's a permit space again.

TANKS & CISTERNS

Water, fuel and coolant tanks. Cisterns can hold low oxygen from rust or rot.

HULL VOIDS

Spaces between hab walls and decks. Often purged with inert gas.

DUCTS & PIPELINES

Air, water and service runs big enough to crawl through.

POLYMER VATS

Curing vapors, heat from the cure, and material that can trap you.

SEALED HABS

Unfinished modules with no air system running yet.

SAND HOPPERS

Flowing sand engulfs a person in seconds.

VAULTS & PITS

Utility vaults, sumps and pits. Heavy gases settle here.

TRENCHES & SHAFTS

Open trenches follow HS-06, using the HS-SA air limits. Shafts, tunnels, and any trench with a hazardous atmosphere or an engulfment risk outside the shield are permit-required spaces (HS-SA).

TEST THE AIR FIRST

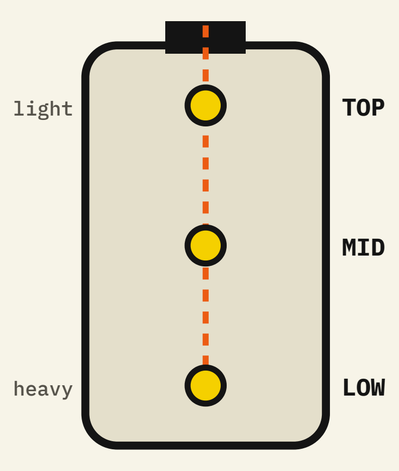
ENTRY LIMITS · BEFORE ENTRY, THEN CONTINUOUSLY

1 · O_2 OXYGEN
19.5–23.5%
Normal air is 20.9%. Under 19.5%, something has pushed the oxygen out. Under 16%, judgment fails. Over 23.5%, clothes and hair burn fiercely. OSHA limits.

2 · LEL FLAMMABLE
< 10%
Of the lower explosive limit. At 10% or above, no entry. OSHA limit. Hot work stops at 1% (site rule).

3 · H_2S
< 10 ppm
NIOSH ceiling (10 min) and OSHA construction limit (8 hr). At 100 ppm it is immediately dangerous to life. It kills your sense of smell.

4 · CO
< 35 ppm
NIOSH 8-hour limit, used here as the entry limit (site rule). OSHA limit: 50 ppm. No smell, no color.



1. **Who tests:** the entry supervisor, or a gas tester trained on that meter and named on the PTW-CS. They test from outside the hatch before entry and every hour, and initial each reading on the permit.
2. **Bump-test the meter** before each day's use, and check the calibration sticker. Failed, or out of date? Don't use it.
3. **Oxygen first.** Most meters need enough oxygen to read flammables correctly.
4. **Flammables next.** Gas, fuel vapor, solvent and polymer off-gas.
5. **Toxics last.** H_2S, CO, and whatever the job brings, like welding fumes or resin vapor.
6. **Top, middle and bottom.** Light gases rise. Heavy gases like argon, H_2S and fuel vapor sink. Once inside, entrants keep their personal monitor on and probe ahead as they go deeper. Let each reading settle: about 2 seconds per foot of hose, plus the sensor time on the meter.
7. **Keep monitoring.** Every entrant wears a running meter. Entrants may watch any test and ask for a retest.
8. **It alarms: everyone gets out.** No arguing. The attendant tells the entry supervisor and the permit is suspended. Nobody goes back in until the cause is found, the space is ventilated and retested, and the entry supervisor signs the permit again. Treat every alarm as real.

INERT GAS

Argon and nitrogen are used to purge tanks, hull voids and weld zones. They push oxygen out.

ONE BREATH CAN DROP YOU.

In air with almost no oxygen, people collapse within seconds, often with no warning at all. Your body senses rising CO_2. It can't sense missing oxygen, so you never feel short of breath.

- A **purge tag** on a hatch means inert gas has been used inside. Never enter without a fresh test.
- **No tag proves nothing.** Tags fall off, and gas leaks in from next-door voids. Test every hatch, every time.
- **Disconnect inert purge lines** before entry. A closed valve alone doesn't count.
- **Ventilate** with forced fresh air until every level reads normal air (about 20.9%), and keep the fan running. Can't get it to 20.5% or more? Find out why before anyone goes in (site rule).
- Argon is **heavier than air**. It pools in low spots, pits and the bottom of tanks.
- Never put your head into a hatch "just to look". That's the breath that kills.

FOUR ROLES, EVERY ENTRY

NAMED ON THE PERMIT BEFORE ANYONE GOES IN
ROLES: REF-GL

ENTRANT

BADGE ENDORSEMENT

Goes inside. Wears a full-body harness and a running meter. Knows the hazards and the signs of trouble. Answers every check-in. Gets out the moment the attendant says so, the meter alarms, or they feel wrong.

ATTENDANT

BADGE ENDORSEMENT

Stays outside at the hatch the whole time and **never enters**. Does no other job (a fire watch is a separate person). Keeps the sign-in count of who is inside. Checks in with each entrant at least every 5 minutes (site rule). Orders everyone out if the meter alarms, an entrant acts strange, something outside threatens the space, or the attendant can't do the job. Keeps out anyone not on the permit. Calls rescue and starts the retrieval winch. Needs to leave? Everyone comes out first, or a trained attendant takes over by name.

ENTRY SUPERVISOR

FOREMAN OR GENERAL FOREMAN (WHITE HAT) · HS-SA ENDORSEMENT

Checks the tests, the isolation and the rescue plan. Confirms the rescue cover before entry. Signs the PTW-CS. Suspends it after any alarm. Ends it when the job is done, the shift ends, or conditions change.

RESCUE TEAM

FIRE & RESCUE · RED HELMET · CALL SIGN "RESCUE [N]"

Trained and practiced on this type of space. Waits at the hatch or stays on call, as the permit says (see below). Called on Ch 1 with "Confined space".

RESCUE COVER

SET BEFORE THE PERMIT IS SIGNED · WRITTEN ON THE PERMIT

COVER	WHEN	WHAT IT MEANS
Team at the hatch	Inert gas used in or piped into the space. The first air test read outside the limits, even if ventilation fixed it. Engulfment risk (sand, liquid, polymer). Hot work inside. Inside heat at Extreme (125°F) or higher. Non-entry retrieval can't be used.	A Fire & Rescue team waits at the hatch, suited and ready to go in, the whole time anyone is inside.
On-call team	Every other permit entry: air within limits, hazards isolated, non-entry retrieval in place.	Before entry, the entry supervisor confirms a Fire & Rescue sector team is free and can reach the hatch and start rescue within **10 minutes** (site rule). The time goes on the permit.

- Safety & Health evaluates each Fire & Rescue team for each type of space: can they get there in time, with the right gear, and have they practiced a rescue on that type of space in the last 12 months? (Old Earth: 1910.146(k), 1926.1211.)
- Rescue team called away? They tell the entry supervisor at once, and everyone comes out until cover is back.

10/16

In one NIOSH study, 10 of the 16 people who died in confined spaces were would-be rescuers (NIOSH Alert 86-110). They saw a coworker collapse and went in after them. The same air dropped them too. **Don't go in. Pull them out with the line, and call rescue.**

RESCUE WITHOUT GOING IN

1. Every entrant wears a **full-body harness**. The **retrieval line** clips to the back D-ring between the shoulders.
2. For vertical entries over 5 feet deep, the line runs to a **tripod or davit with a mechanical winch**.
3. Non-entry retrieval is required on every entry, unless the line would add risk or wouldn't help (for example, it would snag on bends or near a mixer). Then the entry supervisor writes another rescue method on the permit, and a Fire & Rescue team waits at the hatch.
4. Entrant down, or no answer after two calls? The attendant calls **Ch 1: "Emergency, emergency, emergency. Confined space"**, with the location and how many are inside.
5. The attendant cranks them out with the winch from outside.
6. Only Fire & Rescue in supplied air go in if the line can't get them out.

THE PERMIT (PTW-CS)

Signed by the entry supervisor. Posted at the hatch and on the permit board at the structure. Entry stops at 10% LEL. It shows:

- the space and the job
- start and end time: one shift at most (site rule)
- entrants, attendant and entry supervisor by name
- the hazards, and what is locked out and blanked
- the air limits, plus each reading with time and initials
- the heat reading inside and its heat level
- how you'll talk, and the check-in interval
- the rescue cover (at the hatch, or on call with its time) and how to call it
- equipment: meter, fan, harness, winch, intrinsically safe lights and radios
- other permits: PTW-HW for hot work, PTW-HE for Extreme heat entry

It ends when the job is done, the shift ends, or conditions change. Cancelled permits go to Safety & Health and are kept for one year. All permits: REF-PTW.

BEFORE ANYONE ENTERS

- **Lock out** every pump, mixer, conveyor and power feed (HS-07). Electrical isolation: Supplement B (HS-SB).
- **Pipes into the space** (gas, inert purge, resin, water, sand): blank them, disconnect and cap them, or use double block and bleed. A closed valve alone doesn't isolate a pipe, because valves leak.
- **Ventilate** with forced fresh air. Put the fan intake in the shade, upwind, away from exhaust and engines, with a sand filter.
- **Intrinsically safe (IS) gear** where flammables are possible. IS lights, radios and meters are built so they can't make a spark or get hot enough to set off gas. Look for the IS marking.
- **Comms tested at the hatch.** The permit names how you'll talk: voice, a hard-wired line, or radio on the entry talkgroup named on the permit, never a sector channel (REF-RAD).

HEAT INSIDE

A tank in the sun can run far hotter than the air outside. **Measure inside the space**, at the work spot, before entry and every hour. Use that reading for the heat level and the WBGT work/rest table in HS-03. Follow whichever is more restrictive. A sealed suit or coverall adds the clothing adjustment from HS-03 before you read the table.

INSIDE	RULE
Up to 124°F	HS-03 heat level and work/rest table, from the inside reading.
125–139°F	**Extreme.** HS-03 Extreme rules. Rescue team at the hatch.
140–159°F	**Extreme heat entry.** Needs a **PTW-HE** (signed by the area general foreman, countersigned by a Safety officer) as well as the PTW-CS. Each entry is never over 15 min, then at least 30 min in a cooled space. Standby rescuer in full gear. A medic checks each entrant after every entry: pulse over 120 one minute after stopping, or temperature over 101.3°F, ends that worker's entries for the shift. Only workers past the ramp and Days 6–14, cleared for heat entry that shift by Revival Med Bay or the FAS, may enter (site rule).
160°F+	**Site closed.** No entry except rescue.

HOT WORK INSIDE

- **Needs a PTW-HW** (HS-05) as well as the PTW-CS. The fire watch is a separate person from the attendant.
- **No hot work at 1% LEL or above** (site rule; the entry limit is 10%). Keep a meter running at the work the whole time.
- **Watch the oxygen.** Shielding gas (argon, CO_2) pushes oxygen out. A leaking oxygen hose makes the air oxygen-rich, and clothes burn like paper. Stop and get out if O_2 goes under 19.5% or over 23.5%.
- **Cylinders and welding machines stay outside.**
- **Every break:** close the torch valves, shut the gas off at the cylinders outside, and pull the torch and hoses out of the space (site rule; Old Earth requires removal at shift end).
- Ventilate with fresh air. Never with oxygen.

DO

- Read the permit before you go in.
- Sign in and out with the attendant.
- Watch your meter and your partner.
- Get out the first time something feels off: dizzy, headache, confused, out of breath.
- Tell the attendant every time you change position.

DON'T

- Enter without a test, a permit and an attendant.
- Use pure oxygen to "freshen" the air. It makes fires explosive.
- Unclip your retrieval line to reach further.
- Go in after someone who collapsed.

STOP-WORK AUTHORITY

Anyone can stop unsafe work. No penalty, no questions. Stop it, make it safe, tell your crew lead (your foreman if there's no crew lead; any green hat if it isn't fixed). Work restarts only when the crew lead confirms the fix with the person who stopped it. Every stop is logged.

WORKER ACKNOWLEDGEMENT

I will not enter a confined space without a signed PTW-CS, a fresh air test, a harness on a retrieval line (or the rescue method the permit names) and an attendant outside. I will never go in after someone who collapsed. I hold the badge endorsement for my role before I take it.

NAME ____________________
COLONIST ID ____________________
CREW / SECTOR ____________________
DATE ____________________

Adapted from Old Earth standards: OSHA 29 CFR 1910.146 and 1926 Subpart AA (permit-required confined spaces, incl. 1910.146(k) and 1926.1211 rescue), 1926 Subpart P (excavations), 1910.252(b)(4) and 1926.353 (welding in confined spaces), 1926.55 (construction exposure limits); NIOSH Pocket Guide (H_2S and CO limits); NIOSH Alert 86-110; OSHA SHIB 09-30-2004 (gas monitor calibration); NFPA 51B (hot work).

HS-SA · Dunewright Colony Works · Replace when faded